

Lab-3

1. What is SMPS? Explain its operation.

SMPS (Switched Mode Power Supply) is an electronic power supply used in computers to convert AC mains voltage into regulated DC voltages required by various computer components.

Functions of SMPS

- Converts AC power (230V/110V) into DC power.
- Provides different output voltages such as +3.3V, +5V, and +12V.
- Protects the computer from voltage fluctuations and overloads.
- Supplies stable power to the motherboard, CPU, storage devices, and peripherals.

Operation of SMPS

- a. AC Input
AC power from the mains enters the SMPS.
- b. Rectification and Filtering
The AC voltage is converted into DC using a rectifier and filter circuit.
- c. High-Frequency Switching
The DC is switched on and off rapidly by power transistors at high frequency.
- d. Transformation
The high-frequency signal is passed through a transformer to step up or step down the voltage as required.
- e. Secondary Rectification
The transformed AC signal is converted back into DC.
- f. Voltage Regulation
Feedback circuits monitor the output voltage and adjust the switching process to maintain a stable output.

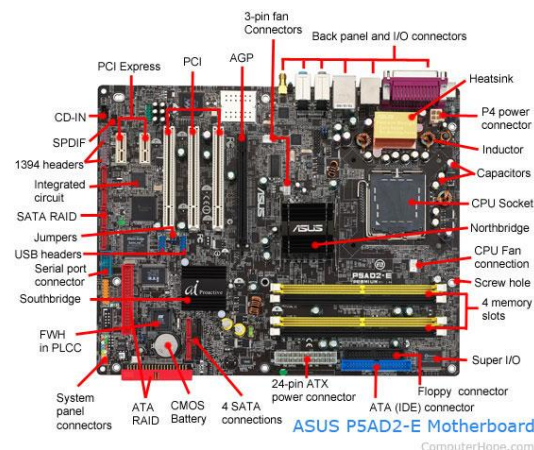
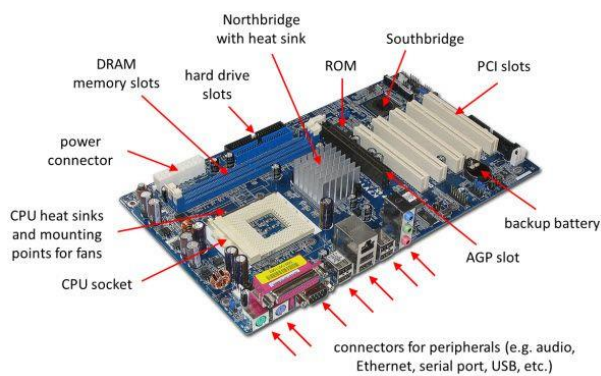
2. Explain the various components available on the mother board.

A motherboard is the main circuit board of a computer that connects and allows communication among all hardware components.

Major Components of a Motherboard

- a. CPU Socket
 - Holds the processor (CPU).
 - Provides electrical connections between the CPU and motherboard.
- b. Chipset
 - Controls data flow between CPU, memory, storage, and peripherals.
 - Modern systems integrate many chipset functions into the CPU.
- c. RAM Slots (DIMM Slots)
 - Used to install memory modules.
 - Temporary storage for running programs and data.
- d. BIOS/UEFI Chip
 - Contains firmware that initializes hardware during startup.
 - Stores system-level instructions.
- e. CMOS Battery
 - Powers the CMOS memory when the computer is off.

- Retains date, time, and BIOS settings.
- f. Expansion Slots
- Examples: PCIe slots.
 - Used for graphics cards, network cards, sound cards, etc.
- g. Storage Connectors
- SATA ports for HDDs and SSDs.
 - M.2 slots for NVMe SSDs.
- h. Power Connectors
- Connect the SMPS to the motherboard.
 - Supply power to the CPU and other components.
- i. USB Headers and Ports
- Allow connection of USB devices.
 - Support keyboards, mice, storage devices, etc.
- j. Audio and Network Controllers
- Provide sound and network connectivity.
 - Often integrated into the motherboard.
- k. Heat Sinks and VRM
- Regulate and distribute power to the CPU.
 - Help dissipate heat.
- l. Input/Output (I/O) Ports
- USB ports, Ethernet port, audio jacks, HDMI/DisplayPort, etc.
 - Used for external device connections.



3. Explain the CMOS set up and ROM BIOS

CMOS Setup

CMOS (Complementary Metal-Oxide Semiconductor) is a small memory chip that stores system configuration settings.

Information Stored in CMOS

- System date and time.
- Boot sequence.

- Hard disk configuration.
- CPU and memory settings.
- Security passwords.
- Power management settings.
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CMOS Setup Utility

- Accessed by pressing keys such as Delete, F2, or Esc during startup.
- Allows users to view and modify hardware settings.

Features of CMOS Setup

- Set system clock and date.
- Configure storage devices.
- Set boot priority.
- Enable or disable onboard devices.
- Configure security options.

ROM BIOS

ROM BIOS (Read Only Memory – Basic Input/Output System) is firmware stored on a ROM/Flash memory chip on the motherboard.

Functions of ROM BIOS

- a. POST (Power-On Self-Test)
Checks hardware components such as RAM, CPU, keyboard, and storage devices during startup.
- b. Hardware Initialization
Initializes and configures hardware devices.
- c. Bootstrap Loader
Loads the operating system from the boot device into memory.
- d. Basic Input/Output Services
Provides low-level routines for keyboard, display, storage, and other devices.

Relationship Between CMOS and BIOS

- BIOS is the firmware program that starts the computer.
- CMOS stores the configuration settings used by the BIOS.
- BIOS reads the CMOS settings during startup and configures the hardware accordingly.